

PHYS 570: Fusion Categories for Physics

Fall 2025 Syllabus

Course Information

- PHYS 570: Fusion Categories for Physics Section 017 CRN 28447
- Tuesday/Thursday 9:00AM - 10:15AM in PHYS 234
- Lecture course (in-person), 3.0 Credit Hours
- Prerequisites: proof-based linear algebra; quantum mechanics
- Brightspace: for announcements and grades only
- Course webpage: <https://www.math.purdue.edu/~colleend/phys570>

Instructor Contact Information

- Colleen Delaney
- Mathematical Sciences Building 440
- (765)-494-0846
- colleend@purdue.edu
- Office Hours: Thursday 2:00PM - 3:00PM in MATH 440 or by appointment
 - Regularly scheduled office hours may be subject to change based on student/instructor availability

Course Description

An introduction to fusion categories and their applications to physics. Students will learn to work with fusion categories and their algebraic precursors with an emphasis on graphical and computational methods with the goal of preparing for research in theoretical/mathematical physics. Applications to be covered may include but are not limited to: anyon models and topological phases of matter in (2+1)D, topological quantum field theory, generalized symmetries in (1+1)D, quantum error correction, topological quantum computation, superselection theory, and (1+1)D conformal field theory.

Resources, Technology & Texts

- The course webpage <https://www.math.purdue.edu/~colleend/phys570> will be the main resource for the course material outside of class.
- The course is designed to be self-contained from the instructor's lectures and lecture notes.
- Lecture notes will be posted to the course webpage following each class.
- We will not be following or requiring any one textbook or set of notes as the content will draw from many different primary sources in different fields. Supplemental open-access reading may be suggested for each unit/topic as the course progresses.
- Students are encouraged to attend instructor office hours to discuss exercises or questions related to the course.

Learning Outcomes

1. Perform calculations in elementary group theory and representation theory and appreciate the role they play in physics.
2. Learn the definition of a skeletal unitary (and modular) fusion category and be able to work with the graphical calculus on trivalent graphs.
3. Learn the abstract definition of a fusion, braided fusion, modular, and module category and be able to manipulate string diagrams.
4. Understand the role of fusion categories in the theory of anyons, topological quantum field theory, topological quantum computation, generalized symmetry, and quantum error correction.
5. Be able to read recent research articles in theoretical physics or mathematics that use fusion category methods and communicate their results.

Grade Breakdown

Students may take the course for a letter grade or Pass/Fail. Grades will be assigned based on the following assessments and are designed to measure student achievement of the learning outcomes given above.

Assessments	Timeline	Percentage of Total Grade
Participation	Throughout semester	10%
Exercises	Collected biweekly in class	60%
Final Project	To be presented or collected December 2025 in class	30%

Participation

Students who make a good faith effort to participate in the course and who volunteer twice to photograph the chalkboards during lecture will receive full credit on this portion of their grade. (The instructor will provide a device if the student does not wish to use a personal device for any reason.)

Here are some other possible ways to participate in the course:

- Regular attendance
- Asking questions during lecture or office hours
- Providing feedback through an optional mid-semester survey

Exercises

Exercises will be posted on the course website and collected approximately every other week at the beginning of class on Thursdays starting Thursday September 11, 2025. (See the course schedule for all collection dates.)

- Exercise solutions can be either handwritten or typed.

- Collaboration is allowed in solving homework problems but each student is responsible for writing up and submitting their own solutions; no one should submit solutions they cannot independently reproduce.
- If you are not able to submit your homework in class on any given Thursday you may:
 - slide it under the instructor's office door in MATH 440 or
 - turn it in to the instructor's mailbox in MATH 427.
- Exercises will be reviewed by our graduate student grader for completeness/understanding and will be returned in class within 10 days of submission.
- Students needing more time to complete exercises may request an extension by emailing the instructor ahead of the Thursday deadline.

Final Project

Starting in November students will select a research article related to the course.

- The instructor will provide a list of suggestions for articles and can make recommendations to students based on their background and interests.
- Students are also welcome to propose their own article subject to instructor approval.

In December, students will present an executive summary of their research article through an in-class presentation. Students will be asked to provide constructive feedback on one another's presentations with regards to clarity of communicating concepts.

Exams

There will be no exams in this course.

Course Schedule

The course schedule can be found on the course webpage; exercise collection dates will also be posted in Brightspace.

Academic Integrity

Students are expected to abide by the Purdue Honor Pledge:

"As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - We are Purdue."

Parameters and Boundaries for the Use of Artificial Intelligence

It is against the spirit of this course to prompt generative AI/LLM models with the exercises assigned in this course in an attempt to produce solutions in place of solving them independently. Students who do not abide by this policy will not receive credit for the offending portion of the assignment.

Students may not submit AI/LLM-generated text as part of their assigned exercises or final project. Additional guidelines for the use of AI/LLMs will be discussed later in the course when more detailed instructions for the final project are given.

AI/LLMs will not be used by the instructor or grader to evaluate your work or in the preparation of grades.

Grading Scale

Students who earn a percentage of total points in the course greater than or equal to the following values are guaranteed the corresponding letter grade.

- $\geq 97\%$ A+
- $\geq 93\%$ A
- $\geq 90\%$ A-
- $\geq 87\%$ B+
- $\geq 83\%$ B
- $\geq 80\%$ B-
- $\geq 77\%$ C+
- $\geq 73\%$ C
- $\geq 70\%$ C-
- $\geq 67\%$ D+
- $\geq 63\%$ D
- $\geq 60\%$ D-

Attendance Policy

- Please stay home if you are feeling unwell and please consider wearing a mask if attending lecture while experiencing symptoms of a contagious respiratory illness.

Nondiscrimination Statement

Purdue University is committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach their potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. A hyperlink to Purdue's full Nondiscrimination Policy Statement is also included in our course Brightspace under University Policies and Statements.

Accessibility

Students that can benefit from accommodations provided from the Disability Resource Center are encouraged to connect with them by email at drc@purdue.edu or by phone at (765)-494-1247 at the beginning of each semester.

Students are responsible for ensuring that the instructor is provided with a Course Accommodation Letter and are encouraged (but not required) to contact the instructor during the early in the semester to ensure that accommodations are provided to the best of the instructor's ability.

If students encounter course materials (notes, video, transcripts, webpages, supplementary documents) or processes which are not accessible or fair please notify the instructor so that an accommodation can be provided.

Copyright Statement

Students may not post the instructor's course materials (or any work in which they are not the sole or joint author or have not obtained the permission of the authors) to commercial websites, media sources, or other outlets (e.g., Chegg, Course Hero, Reddit, Twitter, YouTube etc.). Students who do not abide by this policy will not receive credit for the offending assignment and may be subject to additional university policy.

Mental Health/Wellness Statement

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS). TAO is available to all students at any time by creating an account on the [TAO Connect website](#), or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the [Office of the Dean of Students](#). Call 765-494-1747. Hours of operation are M-F, 8 a.m.- 5 p.m.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a [Purdue Wellness Coach at RecWell](#). Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact [Counseling and Psychological Services \(CAPS\)](#) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

Emergency Preparedness and Responsiveness

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be announced via email to your @purdue.edu address and on our Brightspace/course webpage. The version of the syllabus made available through our Brightspace/course webpage will also be updated accordingly.

A link to Purdue's Information on Emergency Preparation and Planning is located on our Brightspace under "University Policies and Statements." Classroom-specific instructions for responding to an emergency taking place during our class time will be reviewed briefly on the first day of class per Purdue's required Emergency Preparedness Briefing.